

A note on a question about polynomial multiplicities over $\mathbb{F}_2$

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Abstract

Bishnoi, Boyadzhyska, Das and Mészáros ask whether a polynomial over $\mathbb{F}_2$ that vanishes with multiplicity at least k at every nonzero point of $\mathbb{F}_2^n$, and with multiplicity at most $k - 1$ at the origin, has degree at least $2k - \lfloor k/2^{n-1} \rfloor$, when $k \geq 2^{n-2}$. We observe that the requested inequality is an immediate consequence of the multiplicity Schwartz–Zippel lemma of Dvir, Kopparty, Saraf and Sudan. The deduction works for all positive integers n, k , without the stated restriction on k .

1 The question and the standard estimate

The question below can be answered by adding up the required multiplicities. The standard estimate bounds their sum by the polynomial's degree times half the size of the binary grid. Comparing these two totals gives the requested degree bound.

Let $\mathbb{F}$ be a field and $n \geq 1$. All degrees below are ordinary total degrees of formal polynomials. For a nonzero polynomial $P \in \mathbb{F}[x_1, \dots, x_n]$ and $a \in \mathbb{F}^n$, write

$$P(a + z) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} c_\alpha z^\alpha.$$

Its multiplicity at a is

$$\text{mult}_a(P) = \min\{|\alpha| : c_\alpha \neq 0\}, \quad |\alpha| = \alpha_1 + \dots + \alpha_n.$$

This is Hasse multiplicity: all Hasse derivatives of total order less than $\text{mult}_a(P)$ vanish at a . We set $\text{mult}_a(0) = \infty$. In particular, a nonvanishing value has multiplicity zero. We do not reduce polynomials modulo the field equations, since that operation need not preserve multiplicities.

In the published version of *Subspace coverings with multiplicities*, Bishnoi, Boyadzhyska, Das and Mészáros [1, Question 4.3] ask the following: for positive integers n, k with $k \geq 2^{n-2}$, if $P \in \mathbb{F}_2[x_1, \dots, x_n]$ satisfies

$$\text{mult}_a(P) \geq k \quad (a \in \mathbb{F}_2^n \setminus \{0\}), \quad \text{mult}_0(P) \leq k - 1, \tag{1}$$

must one have

$$\deg P \geq 2k - \left\lfloor \frac{k}{2^{n-1}} \right\rfloor? \tag{2}$$

We use the journal numbering; the corresponding question in the earlier arXiv version is [2, Question 5.2].

The relevant estimate is already available in the work of Dvir, Kopparty, Saraf and Sudan [3, Definition 2.2 and Lemma 2.7].

Lemma 1 (Multiplicity Schwartz–Zippel). *Let $\mathbb{F}$ be any field, $n \geq 1$, and let $P \in \mathbb{F}[x_1, \dots, x_n]$ be a nonzero polynomial of total degree d . For every finite nonempty set $S \subseteq \mathbb{F}$,*

$$\sum_{a \in S^n} \text{mult}_a(P) \leq d |S|^{n-1}. \quad (3)$$

There is no restriction relating d to the field size in this lemma.

2 The deduction

Proposition 1. *For every pair of positive integers n, k , every polynomial $P \in \mathbb{F}_2[x_1, \dots, x_n]$ satisfying (1) obeys*

$$\deg P \geq 2k - \left\lfloor \frac{k}{2^{n-1}} \right\rfloor.$$

Proof. The condition at the origin excludes the zero polynomial. Put $d = \deg P$ and apply Lemma 1 with $S = \mathbb{F}_2$. Since multiplicities are nonnegative and $\mathbb{F}_2^n$ has $2^n - 1$ nonzero points,

$$k(2^n - 1) \leq \sum_{a \in \mathbb{F}_2^n} \text{mult}_a(P) \leq d 2^{n-1}.$$

Dividing by 2^{n-1} and using integrality of d gives

$$d \geq \left\lceil 2k - \frac{k}{2^{n-1}} \right\rceil = 2k - \left\lfloor \frac{k}{2^{n-1}} \right\rfloor.$$

□

Proposition 1 implies the bound asked for in the published Question 4.3 of [1] without the assumption $k \geq 2^{n-2}$. The upper bound on the origin’s multiplicity is used here only to ensure that $P \neq 0$. In fact, the same argument applies to any nonzero polynomial with the required multiplicities at the other points.

This direct application of the standard lemma extends the range of validity, without asserting sharpness. For $k = 1$ and $n \geq 3$, it gives only $\deg P \geq 2$, whereas Alon–Füredi gives $\deg P \geq n$ over any field [4, Theorem 5].

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References

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